Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-linear representations in quantum mechanics

Files.

- File: `briefs/NonLinearQuantum/NonLinearQuantum.lean`
- Library: `NonLinearQuantum`
- Imports: `Mathlib.Analysis.InnerProductSpace.Calculus`, `Mathlib.Analysis.SpecialFunctions.ExpDeriv`

Setting. $H = \mathbb{C}^2$ (`EuclideanSpace ℂ (Fin 2)`), with $|z^+\rangle = e_0$, $|z^-\rangle = e_1$, and

$$m(v) = e^{i\theta\, p(z^+|v)}\, v, \qquad p(z^+ \mid v) = \frac{|v_0|^2}{\|v\|^2}.$$

In Lean $m(0) = 0$ holds without a case split, since $0/0 = 0$.

Results.

Brief	Lean
eq. (3), rays to rays	`nlMap_smul`: $m(kv) = k\,m(v)$ for every $k \in \mathbb{C}$
each ray is mapped to itself	`nlMap_mem_ray`
eq. (4), m not linear	`nlMap_not_add`, `not_isLinearMap_nlMap`, for $\theta \in (0, 2\pi)$
eq. (5), inner product	`inner_nlMap`
eq. (6), Born rule preserved	`born_nlMap`
Remark: m bijective	`nlHomeomorph`: a homeomorphism of $\mathbb{C}^2$, inverse $m_{-\theta}$
Remark: m a diffeomorphism (correction)	`contDiffAt_nlMap`: smooth at every $v \neq 0$
	`not_differentiableAt_nlMap_zero`: not differentiable at 0
Theorem 1, mathematical content	`exists_nonlinear_born_preserving`

Differences from the brief. Theorem 1 ("REP-VSLIN is independent of QST") is a statement about the logical independence of two conditions. The checked content is the existence of a homeomorphism of H that maps every ray to itself, preserves the Born rule and is not additive.

Problems found in the brief. The closing remark says that m is a diffeomorphism. It is not differentiable at 0: along every real line $m(tv) = t\,m(v)$, so a derivative at 0 would have to equal m itself, and m is not additive. The correct statement is that m is a homeomorphism of $\mathbb{C}^2$ and a diffeomorphism of $\mathbb{C}^2 \setminus \{0\}$. Theorem 1 only needs a bijection, so it is not affected.